

Handoff/Closeout Documentation

Project name: Ava - Personal AI for Smarter Healthcare

Sponsor organization: Photon

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Project duration: 6 months

Purpose of the Project

This project aims to create a healthcare support agent that works through iMessage to help users better understand and manage medication-related instructions. The product focuses on making medication routines easier to follow by allowing users to upload prescription documents or medication instructions, then receive structured reminders and supportive meal-related guidance.

The current project focuses on three main goals:

1. Help users parse medication instructions from documents, images, or text.
2. Generate medication reminder schedules based on dosage, frequency, duration, and meal timing.
3. Provide food and meal guidance that supports safe medication intake, such as identifying whether a medication should be taken with food or separated from certain ingredients.

The long-term goal is to make health instructions more accessible, especially for users who may feel overwhelmed by medical language, complex schedules, or multiple medications

Target Users

1. Primary Users: Medication-taking patients (people who need help understanding and following medication instructions)
 - a. Take one or more medications regularly
 - b. Receive confusing prescription instructions
 - c. Need reminders for when and how to take medicine
 - d. Want simple explanations of medication timing, food requirements, or restrictions
 - e. Prefer texting over using a separate medication app
2. Secondary Users: Caregivers and family members (use the agent to help someone else manage medication routines)
 - a. Adult children helping parents
 - b. Parents managing a child's medication
 - c. Partners supporting each other's care
 - d. Informal caregivers who help track medication schedules.
3. Tertiary Users: People with food-medication concerns (people who need help understanding what food they might need to avoid while taking specific medicine)

Stakeholders

1. Patients / End users
2. Caregivers
3. Healthcare professionals
4. Project sponsor / project owner
5. Development team
6. Data/API providers (RxNav, DailyMed, USDA FoodData Central)

Relevant Research

- Background Research Conducted: across three areas
- Medication adherence barriers among patients managing prescriptions independently, food-drug interaction awareness and gaps in existing tools, and conversational AI design patterns for health guidance contexts
- Literature review was supplemented by an internal user survey and contextual inquiry
- Key Findings:
 - Medication non-adherence is driven by confusing instruction language and lack of reminders
 - Food-drug interactions are underestimated by patients and underserved by existing tools
 - Plain-language re-framing of prescription instructions improves follow-through
 - iMessage shows higher engagement than app-based health tools among 18-25 year olds
 - AI health tool trust depends on clear scope and easy access to a human provider
- References/Sources:
 - RxNav/RxNorm API (U.S. National Library of Medicine)
 - DailyMed API (FDA drug label data)
 - USDA FoodData Central
 - NIH MedlinePlus

Design and UX Direction

Our design direction focused on making Ava feel more conversational, supportive, and easy to follow for users who may be overwhelmed after a doctor's visit. Since our target users may have low health literacy or may be managing a chronic condition for the first time, we wanted the interaction to feel less like a technical chatbot and more like a guided assistant.

We improved the user experience in several ways. First, we added a typing animation while Ava is processing the user's message. This gives users a clearer sense that the agent is actively

working, especially when it is extracting prescription details, generating a meal plan, or preparing a PDF.

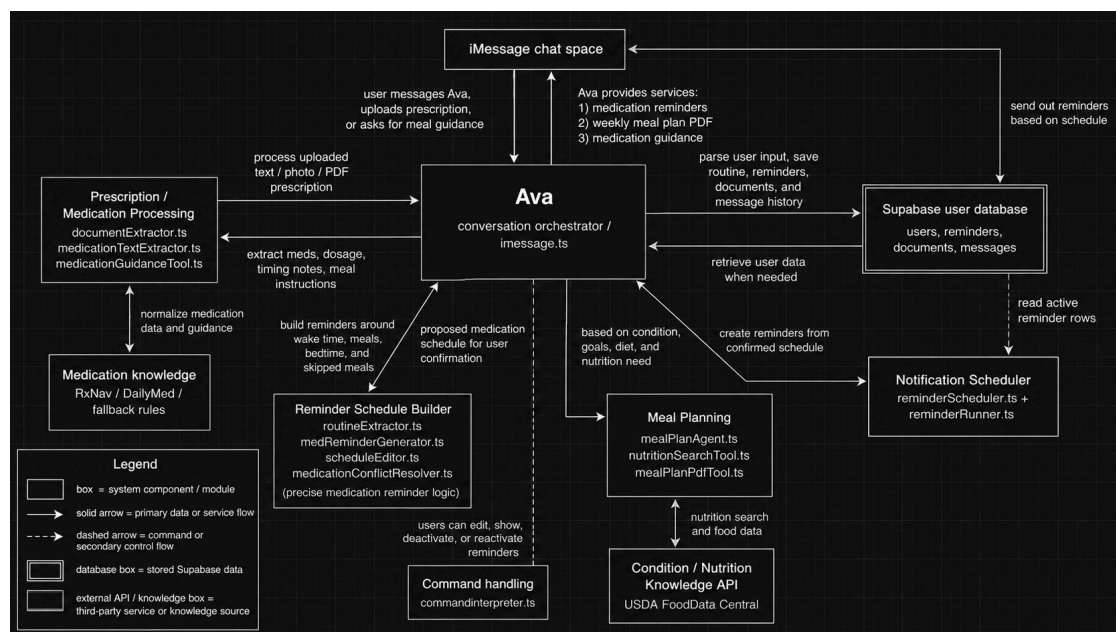
Second, we redesigned the question-asking flow so Ava does not ask random or repetitive questions. Instead, the agent now asks questions in a more reasonable order based on what information is missing. For example, before generating reminders, Ava collects the user's wake time, bedtime, meal schedule, and skipped meals. This makes the conversation feel more natural and helps Ava create reminders that match the user's daily routine.

Third, we created an HTML template for the generated meal plan content. Instead of sending plain text only, Ava can place the generated content into a structured layout before converting it into a PDF. This makes the final output more polished, readable, and easier for users to follow.

We also refined the medication reminder content. The reminders are not just clock-based alerts. They also include useful medication context, such as whether the medication should be taken with food, whether the user should avoid certain foods, or whether there are timing restrictions. This makes the reminders more meaningful and helps users better understand what they need to do when taking each medication

Code Documentation

- Repository location
 - [Agent Github](#)
 - [Information Page Website](#)
 - [Website Github](#)
- Technical architecture



- Setup/installation instructions
 - Prerequisites
 - Node.js >=22.13.0 (the version pinned in engines). Check with `node -v`.
 - Access to a running Photon iMessage server (requires a Mac with iMessage signed in)
 - Accounts/keys for Anthropic, Supabase, and (optionally) USDA FoodData Central
 - Steps
 - Clone and install dependencies
 - `git clone <repo-url>`
 - `npm install`
 - Create a `.env` file in the project root with the variables below (see the Environment Variables section for what each one does):
 - `ANTHROPIC_API_KEY=sk-ant-...`
 - `PHOTON_SERVER_URL=https://your-photon-server`
 - `PHOTON_API_KEY=...`
 - `SUPABASE_URL=https://your-project.supabase.co`
 - `SUPABASE_SERVICE_ROLE_KEY=...`
 - `USDA_API_KEY=...` # optional; falls back to `DEMO_KEY`
 - `.env` is git-ignored. Never commit real keys.
 - Run the assistant
 - `npm run imessage`
 - This launches `src/imessage.ts` via `tsx`, connects to the Photon server, and begins listening for incoming iMessages. There is no separate build step — `tsx` runs the TypeScript directly (`tsconfig.json` is `noEmit`).
- Dependencies
 - Runtime (dependencies)
 - `@anthropic-ai/sdk` — ^0.39.0 — Claude LLM API client
 - `@photon-ai/advanced-imessage-kit` — ^1.14.3 — iMessage send/receive + attachments
 - `@supabase/supabase-js` — ^2.104.0 — Database client
 - `dotenv` — ^17.3.1 — Loads environment variables from `.env`
 - `puppeteer` — ^25.0.4 — Headless Chrome — renders HTML meal plans to PDF
 - Development (devDependencies)
 - `typescript` — ^5.9.3 — TypeScript compiler
 - `@types/node` — ^24.10.9 — Node.js type definitions
 - `@types/pdfkit` — ^0.17.6 — PDFKit type definitions

- Runtime requirements
 - Node.js: $\geq 22.13.0$ (see engines in package.json)
 - Module type: ESM ("type": "module")
 - Run scripts with tsx (e.g. npm run imessage)
- Known issues/limitations
 - Medical accuracy risk
 - The system should not be treated as a replacement for a doctor, pharmacist, or official prescription instructions. Medication and food guidance can be risky if incorrect, so the agent should use cautious wording and recommend professional confirmation when needed
 - Document parsing may be imperfect
 - PDF or image-based prescription documents may have inconsistent formatting. The parser may miss or incorrectly extract information if the document is blurry, incomplete, handwritten, or formatted unusually
 - API dependency limitations
 - RxNav and DailyMed are external services. If their APIs are unavailable, rate-limited, or return incomplete results, medication normalization and label lookup may fail